ReDoS in HeaderUtils::split() Reachable Through the Cache-Control Request Header

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Content

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ReDoS in HeaderUtils::split() Reachable Through the Cache-Control Request Header

- Advisory: GHSA-r925-mgvr-h9cc
- CVE: CVE-2026-108082
- Severity: high
- Published: 2026-10-15
- Updated: 2026-10-15

Affected

- `symfony/http-foundation` (composer): >=4.1.0, <5.4.54
- `symfony/http-foundation` (composer): >=6, <6.4.46
- `symfony/http-foundation` (composer): >=7, <7.4.19
- `symfony/http-foundation` (composer): >=8, <8.1.7
- `symfony/symfony` (composer): >=4.1.0, <5.4.54
- `symfony/symfony` (composer): >=6, <6.4.46
- `symfony/symfony` (composer): >=7, <7.4.19
- `symfony/symfony` (composer): >=8, <8.1.7

Description

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`HeaderUtils::split()` parsed header values with a pattern whose token alternative also matched whitespace, followed by a `(?<!\s)` lookbehind. A run of whitespace placed before a separator could be divided among the iterations of the enclosing repetition in exponentially many ways, and the engine tried all of them before settling. `HeaderBag` calls it for the `Cache-Control` header whenever a `Request` is built, so any application handling HTTP requests is reachable, with no authentication.

A single request carrying such a header costs close to a second of CPU (813 ms for a run of 17 spaces in an 8 KB header on PHP 8.5, doubling with each extra space). From 18 spaces on, `preg_match_all()` gives up at `pcre.backtrack_limit` and `split()` returns an empty array, so every directive of the header is silently dropped.

Resolution

The pattern now consumes whitespace only when more of the value follows it, so a value can no longer end on whitespace, nothing has to be given back, and every quantifier is possessive. The first release of this fix broke splitting on a whitespace separator: `HeaderUtils::split('foo bar', ' ')` returned a single value in 6.4.46 to 6.4.47, 7.4.19 to 7.4.20 and 8.1.7 to 8.1.8. Whitespace characters used as separators are now left out of the whitespace branch, which 6.4.48, 7.4.21 and 8.1.9 contain.

The patches for this issue are available here for branch 5.4 (and forward-ported to 6.4, 7.4, 8.1 and 8.2).

Credits

We would like to thank r3verii for reporting the issue and Nicolas Grekas for providing the fix.

We would also like to thank Bruno Dias from Checkmarx for his help getting a CVE assigned.

References

- Fix: #66048
- Follow-up: #66491

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